

Nitheesh Karthikeyan

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I focus on building solutions that empower people, bridge opportunity gaps, and translate complex systems into accessible, real-world impact.

EDUCATION

VELLORE INSTITUTE OF TECHNOLOGY

B.TECH IN ELECTRONICS AND COMMUNICATION ENGINEERING
2024-2028 | Chennai, India

SBOA SCHOOL AND JR. COLLEGE

2010-2024 | Chennai, India
Active participant and Winner in **Model United Nations**; developed strong skills in structured communication, negotiation, and policy analysis.

CO-CURRICULAR ACTIVITIES

Debate & Public Speaking - Competed in inter-school and college debates; skilled in structured argumentation, policy analysis, and high-pressure communication

Model United Nations (MUN) - Participated in MUNs; engaged in diplomacy, policy negotiation, and collaborative resolution drafting

Tech Leadership & Community Building - Organized campus events with **tech CEOs and industry leaders**; led outreach, partnerships, and execution, impacting **100+ students per event**

PROJECTS

LithoMap — 3D Clinical Research Platform - Built & deployed centralized pipeline at **Government Stanley Medical College**; replaced fragmented FTIR workflows. Standardized spectral dataset enabling early-stage cancer biomarker research; actively used by PG researchers

Quantum-Optimized Reconfigurable Battery System (QAOA, VQE) - Built hybrid quantum-classical optimizer for dynamic battery switching (Qiskit).

Financial Fraud Detection — Quantum Federated Learning (NYU Abu Dhabi) - Built privacy-preserving fraud detection using QVSM + Quantum Differential Privacy. Engineered scalable hybrid pipeline (TensorFlow Federated, Flower, Qiskit, PennyLane)

LANGUAGES

Tamil — Native Proficiency
English - Fluent Public Speaker
Hindi — Professional Working Proficiency
French — Limited Proficiency

REFERENCE

DR. REENA MONICA P.
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Former Dean and current Asst. Director, International Affairs, VIT

WORK EXPERIENCE

MEDXAI RESEARCH SOFTWARE INTERN (WITH STIPEND)

October 2024 - February 2026

- Worked with **20+ pharmaceutical clients** to streamline drug discovery workflows by automating computational processes and narrowing the search space for viable chemical candidates.
- Contributed to the **development of a novel drug candidate** aimed at treating hot flashes during menopause. Successfully passed wetlab, now progressing towards clinical trials.
- Collaborated with interdisciplinary teams across research, healthcare, and engineering to refine solutions aimed at improving real-world health outcomes.

SOCIAL IMPACT & LEADERSHIP

- United Nations Academic Impact — MCN Millennium Fellow (2025)**
 - Selected among global fellows; led sustainability-driven initiatives translating high-level development frameworks into **actionable, student-led programs across 3+ communities**
 - Designed & executed **10+ engagements/workshops**, reaching **200+ students**; enabled grassroots participation in solving local challenges and driving community-level impact
- Paadhai — Career Awareness Initiative**
 - Founded & scaled initiative to bridge career awareness gaps for **government school students from underserved backgrounds**
 - Conducted **monthly sessions impacting 150+ students**; simplified complex career pathways into accessible, actionable guidance, enabling informed academic and career decisions
- Zerodha Varsity — Lead Student Coordinator (Chennai Edition)**
 - Led a **free, large-scale financial literacy initiative** to make investing and personal finance education accessible to students from diverse backgrounds; impacted **500+ participants**
 - Managed **10+ member cross-functional team** and coordinated speakers, content, and operations; delivered a **3-month execution cycle** culminating in a high-engagement event with **INR 3.5L+ prize pool**
 - Focused on breaking financial knowledge barriers by simplifying investing concepts and enabling **first-time learners** to confidently engage with financial systems; **re-invited for external engagements**

HACKATHONS

- IBM Qiskit Fall Fest 2024 — 1st Place Winner.** Presented work to senior leadership (Dr. L. Venkataraman) at IBM Quantum India and reviewed by international faculty.
- IBM Qiskit Fall Fest 2025 — Judge & Teaching Instructor.** Delivered quantum programming sessions to 100+ international participants from 10+ countries; facilitated technical discussions and mentoring in a live and engaging manner.
- Intel+Microsoft SAS Hackathon (HEC Montréal) — World Rank 19.** Optimized business outcomes by analyzing user flow data and applying mathematical models to maximize profits and minimize losses through user-centric, goal-driven strategies.
- Goldman Sachs Quantitative Finance Hackathon — All India Rank 163.** Developed alpha-generating strategies by analyzing portfolio data and applying quantitative models to optimize returns and risk.

AWARDS

- United Nations Millennium Fellow (2025)** — Selected among the top **4%** from over 60,000 applicants for global academic and social impact made through technology on basis of project history.
- 1st Place – Best Startup Idea Award (2025)** presented by Dr. Soumya Swaminathan, former Chief Scientist, World Health Organization.
- Selected as **1 of 5 students** from VIT Chennai for the prestigious **Nokia Campus Connect Program**, featuring rigorous hands-on training in communication systems, machine learning, and computer science.
- Vice President’s Feather-in-the-Cap Award, VIT** — For developing an **ML-based Seat Locator System** deployed in the VIT Library.
- Recognized among the **Top 3 Non-Startup Projects** at VIT Chennai 2024 for developing a **Pothole Detection System** (IIT Madras, ilnventiv)